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%% 1. SYSTEM PARAMETERS (Example 4 - 3rd Order DTS)
A = [0.83, -0.5, 0.02;
      0.1,  0.1, -0.3;
      0.1, -0.08, 0.28];

S = [0.25, -0.01, 0;
      -0.01, 0.1, 0.04;
      0,    0.04, -0.05];

R = eye(3); % R is a 3x3 Identity matrix
Ko = 0; Kq = 1; alpha = 0.01;
exp_term = exp(2*alpha);

% Paper Results (3x3 Matrices)
P_sol = [0.4406, -0.0428, -0.0023;
          -0.0428, 0.4638, -0.0254;
          -0.0023, -0.0254, 0.5217];

J_sol = diag([0.3577, 0.4490, 0.4622]);
E_sol = diag([0.3414, 0.3258, 0.3382]);

%% 2. MATHEMATICAL VERIFICATION
% Constructing the 9x9 composite Phi matrix
Phi11 = Kq^2*A'*J_sol*A - 2*Kq*Ko*A'*E_sol*A - P_sol;
Phi12 = (Ko+Kq)*A'*E_sol;
Phi13 = Kq*A'*J_sol*R - 2*Kq*Ko*A'*E_sol*R - 0.5*A';
Phi22 = exp_term*P_sol - J_sol - 2*E_sol;
Phi23 = exp_term*P_sol*S + (Ko+Kq)*E_sol*R;
Phi33 = exp_term*S'*P_sol*S + R'*J_sol*R - 2*Kq*Ko*R'*E_sol*R - (R+R')/2;

Phi = [Phi11, Phi12, Phi13;
        Phi12', Phi22, Phi23;
        Phi13', Phi23', Phi33];

max_eig = max(real(eig(Phi)));
fprintf('Verification: Max Eigenvalue of 3rd-order Phi is %f\n', max_eig);

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Verification: Max Eigenvalue of 3rd-order Phi is 0.059548

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%% 3. SIMULATION (Trajectories for x1, x2, x3)
k_max = 100;
x = zeros(3, k_max);
x(:,1) = [0.02; 0.01; -0.01]; % Initial Conditions from paper

for k = 1:k_max-1
    % Disturbance w(k) for 3rd order
    if k <= 50
        wk = [0.2*sin(k); 0.14*cos(k); 0.2*cos(k)];
    else

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    wk = [0; 0; 0];
end

yk = A*x(:,k) + R*wk;
fk = max(min(yk, 1), Ko); % Sector [0, 1]
x(:,k+1) = fk + S*wk;
end

%% 4. PLOTTING
figure('Color', 'w');
plot(0:k_max-1, x(1,:), 'r', 'LineWidth', 1.5); hold on;
plot(0:k_max-1, x(2,:), 'g', 'LineWidth', 1.5);
plot(0:k_max-1, x(3,:), 'b', 'LineWidth', 1.5);
grid on;
line([50 50], ylim, 'Color', 'k', 'LineStyle', '--');
title('Example 4: 3rd-Order State Trajectories');
xlabel('Time step k'); ylabel('States x(k)');
legend('x_1(k)', 'x_2(k)', 'x_3(k)');

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